Chapter 9 : Inference for Two-Way Tables

22 June 2024 14:20

- Analysis of a two way table :

EXAMPLE 9.1 No Sweat Labels

Case 8.3 (page 474) concerned the response of men and women to "No Sweat" labels on a garment. These labels supposedly guarantee fair working conditions for the workers who made the garment. Here is the data summary. The table gives the number n of subjects in each gender. The count X is the number who were classified as No Sweat label users.

Population	n	X
1 (women)	296	63
2 (men)	251	27
Total	547	90

To compare men and women, we calculated sample proportions from these counts.

- Two way tables :

⇒ In this chapter we start with a different summary of the same data.

↳ Rather than recording just the count of label users, we record the count of all outcomes in a two-way table.

EXAMPLE 9.2 No Sweat Labels

Here is the two-way table classifying customers by gender and whether or not they are label users:

	Gender		
Label user	Women	Men	Total
Yes	63	27	90
No	233	224	457
Total	296	251	547

Check that this table simply rearranges the information in Example 9.1. Because we are interested in how gender influences label use, we view gender as an explanatory variable and label use as a response variable. This is why we put gender in the columns (like the x axis in a regression) and label use in the rows (like the y axis in a regression). Be sure that you understand how this table is obtained from the table in Example 9.1. Most errors in the use of categorical-data methods come from a misunderstanding of how these tables are constructed.

⇒ We call this particular two-way table a 2×2 table, because there are two rows (Yes & No for label use) and two columns (Women & Men).

↳ The advantage of two-way table is that it can present data for variables having more than two categories by simply increasing the number of rows & columns.

⇒ The question of interest is whether there's a relation between the row variable & the column variable.

↳ ex: Is there a relation between gender & label use?

⇒ We introduce Inference for two-way tables with data that forms 2×2 table. The methodology applies to table in general & will discuss a 3×3 table (two categorical variables each with three categories).

CASE 9.1 **Exclusive Territories and the Success of New Franchise Chains** Many popular businesses are franchises—think of McDonald's. The owner of a local franchise benefits from the brand recognition, national advertising, and detailed guidelines provided by the franchise chain. In return, he or she pays fees to the franchise firm and agrees to follow its policies. The relationship between the local entrepreneur and the franchise firm is spelled out in a detailed contract.

One clause that the contract may or may not contain is the entrepreneur's right to an exclusive territory. This means that the new outlet will be the only representative of the franchise in a specified territory and will not have to compete with other outlets of the same chain. How does the presence of an exclusive-territory clause in the contract relate to the survival of the business? A study designed to address this question collected data from a sample of 170 new franchise firms.[1]

To start our analysis of the relationship between exclusive territories and success we will organize the data in a two-way table. The following example gives the details.

EXAMPLE 9.3 The Two-Way Table

Two categorical variables were measured for each firm. First, the firm was classified as successful or not based on whether or not it was still franchising as of a certain date. Second, the contract of each firm was classified according to whether or not there was an exclusive-territory clause. Here is the 2 × 2 table with the marginal totals:

Observed numbers of firms

	Exclusive Territory		
Success	Yes	No	Total
Yes	108	15	123
No	34	13	47
Total	142	28	170

⇒ The entries in the table of ex 9.3 are the observed, or sample, counts of the number of firms in each category.

↳ For example: there where 108 successful firms with exclusive territory contract, and 15 successful firms without one.

⇒ Were interested in the influence of exclusive territories on success, so exclusive territory (column) is the explanatory variable & success (row) is the response variable.

↳ Each combination of these variables defines a cell.

↳ A two way table with r-rows & c-columns contains r*c cells.

- Describing relation in two-way table:

⇒ To describe relations between categorical variables, we compute & compare percents.

↳ The count in each cell can be viewed as a percent of the grand total, of row total, or column total.

↳ In the first case we are describing joint distribution

Success | Exclusive: Yes | Exclusive: No

Yes | P(Exclusive = Yes & Success = Yes) | P(Exclusive = No & Success = Yes)

↳ In the other case we're examining conditional distribution.

P(Success | Exclusiveness = Yes)

EXAMPLE 9.4 Software Output

CASE 9.1 Figure 9.1 shows the output from Minitab, SPSS, and SAS for the data of Case 9.1. We named the variables SUCCESS and EXCL. The two-way table appears in the outputs in expanded form. Each cell contains four entries. They appear in different orders or with different labels, but all three outputs contain the same information. The frequency or count is the first entry in all three outputs. The row and column totals appear in the margins, just as in Example 9.3. The cell count as a percent of the row total is variously labeled as "% of Row" or "% within SUCCESS" or "Row Pct." For the first cell, this is 108/123, or 87.8%. The cell count as a percent of the column total is also given. You can request yet another entry, cell count divided by the total number of observations (the joint distribution). This is often not very useful and tends to clutter up the output.

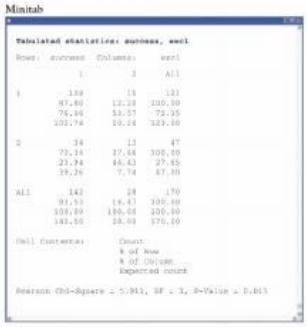
In Case 9.1 we are interested in comparing the two types of firms, those with exclusive territories and those without exclusive territories. The columns of the table contain the information about each firm. Therefore, we examine the column percents to compare the success of the two types of firms. Here they are, rounded from the output for clarity:

Column percents for firms

	Exclusive Territory	
Success	**Yes**	**No**
Yes	76%	54%
No	24%	46%
Total	100%	100%

FIGURE 9.1 Minitab, SPSS, and SAS output for Example 9.4.

Minitab



```
Tabulated statistics: success, excl

Rows: success   Columns: excl

              1         2       All

1           108        15       123
         87.80     12.20    100.00
         76.06     53.57     72.35
        103.74     19.26    123.00

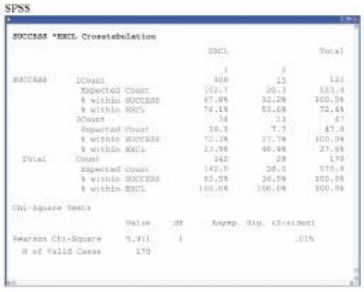
2            34        13        47
         72.34     27.66    100.00
         23.94     46.43     27.65
         39.26      7.74     47.00

All         142        28       170
         83.53     16.47    100.00
        100.00    100.00    100.00
        142.00     28.00    170.00

Cell Contents:      Count
                    % of Row
                    % of Column
                    Expected count

Pearson Chi-Square = 5.911, DF = 1, P-Value = 0.015
```

SPSS



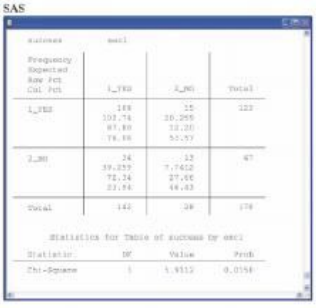
```
SUCCESS * EXCL Crosstabulation

                                        EXCL                Total
                                          1         2
SUCCESS   1Count                         108        15         123
          Expected Count                103.7      19.3       123.0
          % within SUCCESS              87.8%      12.2%      100.0%
          % within EXCL                 76.1%      53.6%      72.4%
          2Count                         34        13          47
          Expected Count                39.3       7.7        47.0
          % within SUCCESS              72.3%      27.7%      100.0%
          % within EXCL                 23.9%      46.4%      27.6%
Total     Count                         142        28         170
          Expected Count                142.0      28.0       170.0
          % within SUCCESS              83.5%      16.5%      100.0%
          % within EXCL                 100.0%     100.0%     100.0%

Chi-Square Tests

                          Value     df     Asymp. Sig. (2-sided)
Pearson Chi-Square        5.911     1                  .015
 N of Valid Cases           170
```

SAS



```
success        excl

Frequency
Expected
Row Pct
Col Pct          1_YES        2_NO       Total

1_YES              108          15         123
                 103.74       19.259
                 87.80        12.20
                 76.06        53.57

2_NO                34          13          47
                 39.259       7.7412
                 72.34        27.66
                 23.94        46.43

Total              142          28         170

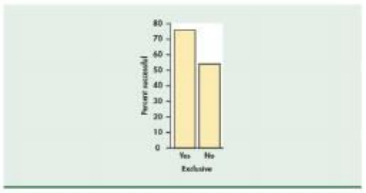
     Statistics for Table of success by excl

Statistic          DF       Value       Prob

Chi-Square          1      5.9112     0.0150
```

The "Total" row reminds us that 100% of each type of firm have been classified as successful or not. (The sums sometimes differ slightly from 100% because of roundoff error.) The bar graph in Figure 9.2 compares the percents. The data reveal a clear relationship: 76% of firms with exclusive territories are successful, as opposed to only 54% of firms that did not have an exclusive-territory contract.

for



- The hypothesis: No association.

⇒ The difference between the percent of successes among the two types of firm is quite large.

↳ A statistical test will tell us whether or not this difference can be plausibly attributed to a random chance.

⇒ The null hypothesis H_0 of interest in a two-way table is: there is no association between row variable & the column variable.

↳ For Case 9.1, null hypothesis state that success & having an exclusive territory are not related.

↳ The alternative hypothesis H_a is that there is an association between these variables.

↳ The alternative H_a doesn't specify any kind of direction for association.

↳ For $5 \times c$ table, the alternative includes many different possibilities, because it includes all the many kinds of association that are possible, we cannot describe H_a as either one-sided or two-sided.

⇒ For $5 \times c$ tables, there are c distributions for the row variable, one for each population.

↳ The null hypothesis then says that the c distributions of the row variable are identical.

↳ The alternative is that the distributions are not all the same.

- Expected cell counts:

⇒ To test the null hypothesis in $5 \times c$ tables, we compare the observed cell counts with expected cell counts calculated under the assumption that null hypothesis is true.

↳ Our test statistic is a numerical measure of the distance between observed & expected cell counts.

EXAMPLE 9.5 Expected Counts from Software

The expected counts for the successful-firms example appear in the computer outputs shown in Figure 9.1. For example, the expected count for the first cell is 102.74.

How is this expected count obtained? Look at the percents in the right margin of the table in Figure 9.1. We see that 72.35% of all firms are successful. If the null hypothesis of no relation between success and exclusive territories is true, we expect this overall percent to apply to both firms with and firms without exclusive territories. In particular, we expect 72.35% of the firms with exclusive territories to be successful. Since there are 142 firms with exclusive territories, the expected count is 72.35% of 142, or 102.74. The other expected counts are calculated in the same way.

⇒ The reasoning of the above example leads to a simple formula for calculating expected cell counts.

↳ To compute the expected count for successful firms with exclusive territories, we multiplied the proportion of successful firms (123/170) by number of firms with exclusive territory (142).

↳ Same way for non-exclusive firm.



success	excl		
Frequency Expected Row Pct Col Pct	1_YES	2_NO	Total
1_YES	108	15	123

	102.74	20.259	
	87.80	12.20	
	76.06	53.57	
2_NO	34	13	47
	39.259	7.7412	
	72.34	27.66	
	23.94	46.43	
Total	142	28	170

Statistics for Table of success by excl

Statistic	DF	Value	Prob
Chi-Square	1	5.9112	0.0150

⇒ from the above figure, we see that the numbers 123 & 142 are the row & column totals for the cell of interest & that 170 is n.

↳ i.e n is total number of observation for the table.

↳ the expected cell count is therefore the product of row total & column total divided by the table total.

Expected Cell Counts

The **expected count** in any cell of a two-way table when the null hypothesis of no association is true is

$$\text{expected count} = \frac{\text{row total} \times \text{column total}}{n}$$

- The chi-squared test:

⇒ we use a statistic that compares entire set of observed counts with the set of expected counts.

↳ first take the difference of each observed count with its corresponding expected count, and then square these values, and divide each squared difference by the expected count.

↳ finally sum over all cells, the result is chi-square statistic X^2.

⇒ If the expected & observed counts are very different then a large value of X^2 will result.

↳ therefore, large value of X^2 provide evidence against the null hypothesis.

Chi-Square Statistic

The **chi-square statistic** is a measure of how much the observed cell counts in a two-way table diverge from the expected cell counts. The recipe for the statistic is

$$X^2 = \sum \frac{(\text{observed count} - \text{expected count})^2}{\text{expected count}}$$

where "observed" represents an observed sample count, "expected" represents the expected count for the same cell, and the sum is over all $r \times c$ cells in the table.



FIGURE 9.3 (a) The $\chi^2(2)$ density curve. (b) The $\chi^2(4)$ density curve.

⇒ Like the t-distribution, the χ^2 distribution form a family describe by a single parameter, the degree of freedom. The figure 9.3 shows $\chi^2(2)$ & $\chi^2(4)$ distributions.

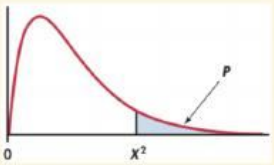
Chi-Square Test for Two-Way Tables

The null hypothesis H_0 is that there is no association between the row and column variables in a two-way table. The alternative is that these variables are related.

If H_0 is true, the chi-square statistic X^2 has approximately a χ^2 distribution with $(r-1)(c-1)$ degrees of freedom.

The P-value for the chi-square test is

$P(\chi^2 \geq X^2)$



where χ^2 is a random variable having the $\chi^2(\text{df})$ distribution with $\text{df} = (r-1)(c-1)$. If the P-value is sufficiently small, we reject the null hypothesis of no association. In this case, we say that the data provide evidence for us to conclude that there is an association.

- The chi-square test always uses the upper tail of the χ^2 distribution.
  - Because any deviation from the null hypothesis makes the statistic larger (least squares)
- The approximation of the distribution of X^2 becomes more accurate as all counts increase.

EXAMPLE 9.6 Exclusive Territories and Franchise Firm Success

The results of the chi-square significance test for the successful-firms example appear in the lower parts of the computer outputs in Figure 9.1, labeled Chi-Square or Pearson Chi-Square. Because all the expected cell counts are moderately large, the χ^2 distribution provides accurate P-values. We see that $X^2 = 5.91$ and $\text{df} = 1$. Examine the outputs and find the P-value in each output. The rounded value is $P = 0.015$. As a check we verify that the degrees of freedom are correct for a 2×2 table:

$$\text{df} = (r-1)(c-1) = (2-1)(2-1) = 1$$

The chi-square test confirms that the data contain clear evidence against the null hypothesis that there is no relationship between success and exclusive territories. Under H_0, the chance of obtaining a value of X^2 greater than or equal to the calculated value of 5.91 is small—less than 15 times in 1000. The test does not tell us what kind of relationship is present. It is up to us to see that the data show that firms with exclusive territories are more likely to be successful. You should always accompany a chi-square test by percents such as those in the table on page 494 and Figure 9.2 and by a description of the nature of the relationship.

The chi-square & the z-test

⇒ We now have two ways to test the hypothesis of equality of two population proportions.

- the chi-square & two sample z-test.
  - In fact these test gives exactly the same result.
    - Because chi-square statistic is equal to the square of z-statistic
      - And $\chi^2(1)$ critical values are equal to square of corresponding $N(0,1)$ critical values.

⇒ The advantage of z-test is that we can test either one-sided or two sided alternative & add confidence intervals to significance test.

⇒ The advantage of chi-square test is that it is more general:
we can compare more than two population proportions or more
generally test - ask about relation in two-way table of any

size.

- Beyond the Basics: Meta-Analysis.

Policymakers wanting to make decisions based on research are sometimes faced with the problem of summarizing the results of many studies. These studies may show effects of different magnitudes, some highly significant and some not significant. What *overall conclusion* can we draw? **Meta-analysis** is a collection of statistical techniques designed to combine information from different but similar studies. Each individual study must be examined with care to ensure that its design and data quality are adequate. The basic idea is to compute a measure of the effect of interest for each study. These are then combined, usually by taking some sort of weighted average, to produce a summary measure for all of the studies. Of course, a confidence interval for the summary is included in the results. Here is an example.

EXAMPLE 9.7 Vitamin A Saves Lives of Young Children

Vitamin A is often given to young children in developing countries to prevent night blindness. It was observed that children receiving vitamin A appear to have reduced death rates. To investigate the possible relationship between vitamin A supplementation and death, a large field trial with over 25,000 children was undertaken in Aceh Province of Indonesia. About half of the children were given large doses of vitamin A, and the other half were controls. The researchers reported a 34% reduction in mortality (deaths) for the treated children who were 1 to 6 years old compared with the controls. Several additional studies were then undertaken. Most of the results confirmed the association: treatment of young children in developing countries with vitamin A reduces the death rate; but the size of the effect varied quite a bit.

How can we use the results of these studies to guide policy decisions? To address this question, a meta-analysis was performed on data from eight studies.[3] Although the designs varied, each study provided a two-way table of counts. Here is the table for the study conducted in Aceh Province. A total of $n = 25,200$ children were enrolled in the study. Approximately half received vitamin A supplements. One year after the start of the study, the number of children who had died was determined.

	Vitamin A	Control
Dead	101	130
Alive	12,890	12,079
Total	12,991	12,209

The summary measure chosen was the **relative risk:** the ratio formed by dividing the proportion of children who died in the vitamin A group by the proportion of children who died in the control group. For Aceh, the proportion who died in the vitamin A group was

$$\frac{101}{12,991} = 0.00777$$

or 7.7 per thousand. For the control group, the proportion who died was

$$\frac{130}{12,209} = 0.01065$$

or 10.6 per thousand. The relative risk is therefore

$$\frac{0.00777}{0.01065} = 0.73$$

Relative risk less than 1 means that the vitamin A group has the lower mortality rate.

The relative risks for the eight studies were

0.73 0.50 0.94 0.71 0.70 1.04 0.74 0.80

A meta-analysis combined these eight results to produce a relative risk estimate of 0.77 with a 95% confidence interval of (0.68, 0.88). That is, vitamin A supplementation reduced the mortality rate to 77% of its value in an untreated group. The confidence interval does not include 1, so we can reject the null hypothesis of no effect (a relative risk of 1). The researchers examined many variations of this meta-analysis, such as using different weights and leaving out one study at a time. These variations had little effect on the final estimate.

- Formulas & Models for two-way tables:

Computations for Two-Way Tables

1. Calculate descriptive statistics that convey the important information in the table. Usually these will be column or row percents.
2. Find the expected counts and use these to compute the X^2 statistic.
3. Use chi-square critical values from Table F to find the approximate P-value.
4. Draw a conclusion about the association between the row and column variables.

CASE 9.2

Background Music and Consumer Behavior Market researchers know that background music can influence the mood and purchasing behavior of customers. One study in a supermarket in Northern Ireland compared three treatments: no music, French accordion music, and Italian string music. Under each condition, the researchers recorded the numbers of bottles of French, Italian, and other wine purchased.[4] Here is the two-way table that summarizes the data:

Counts for wine and music				
	Music			
Wine	None	French	Italian	Total
French	30	39	30	99
Italian	11	1	19	31
Other	43	35	35	113
Total	84	75	84	243

This is a 3×3 table, to which we have added the marginal totals obtained by summing across rows and columns. For example, the first-row total is $30 + 39 + 30 = 99$. The grand total, the number of bottles of wine in the study, can be computed by summing the row totals, $99 + 31 + 113 = 243$, or the column totals, $84 + 75 + 84 = 243$. It is a good idea to do both as a check on your arithmetic.

- Conditional Distributions:-

⇒ First we summarize the observed relation between the music being played and the type of wine purchased.

⇒ The researchers expected that music would influence sales.

↳ music ⇒ explanatory & type of wine ⇒ response.

⇒ In general, the clearest way to describe their relationship is to compare the conditional distributions of the response variable for each value of explanatory variable.

variable. So we will compare the column percents that give the conditional distribution of purchases for each type of music played.

EXAMPLE 9.8 Wine Sales, Given Music Type

When no music was played, there were 84 bottles of wine sold. Of these, 30 were French wine. Therefore, the column proportion for this cell is

$$\frac{30}{84} = 0.357$$

That is, 35.7% of the wine sold was French when no music was played. Similarly, 11 bottles of Italian wine were sold under this condition, and this is 13.1% of the sales.

$$\frac{11}{84} = 0.131$$

In all, we calculate nine percents. Here are the results:

Column percents for wine and music				
	Music			
Wine	None	French	Italian	Total
French	35.7	52.0	35.7	40.7
Italian	13.1	1.3	22.6	12.8
Other	51.9	46.7	41.7	46.5
Total	100.0	100.0	100.0	100.0

In addition to the conditional distributions of types of wine sold for each kind of music being played, the table also gives the marginal distribution of the types of wine sold. These percents appear in the rightmost column, labeled "Total."

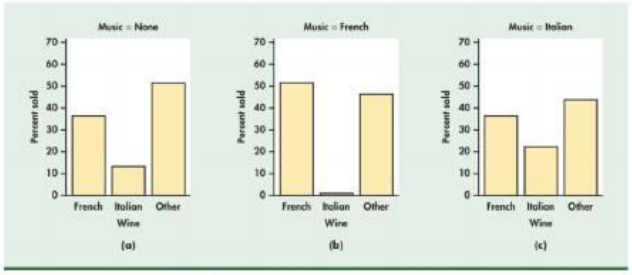


FIGURE 9.4 Comparison of the percents of different types of wine sold for different music conditions, for Example 9.8.

type of wine that customers buy. Sales of Italian wine are very low when French music is playing but are higher when Italian music or no music is playing. French wine is popular in this market, selling well under all music conditions but notably better when French music is playing.

Another way to look at these data is to examine the row percents. These fix a type of wine and compare its sales when different types of music are playing. Figure 9.5 displays these results. We see that more French wine is sold when French music is playing.

Similarly for Italian wine. The negative effect of French music on sales of Italian wine is dramatic.

We observe a clear relationship between music type and wine sales for the 243 bottles sold during the study. The chi-square test assesses whether this observed association is statistically significant, that is, too strong to occur often just by chance. The test only confirms that there is some relationship. The percents we have compared describe the nature of the relationship. What is more, the chi-square test does not in itself tell us what population our conclusion describes. If the study was done in one market on a Saturday,

the results may apply only to Saturday shoppers at this market. The researchers may invoke their understanding of consumer behavior to argue that their findings apply more generally, but that is beyond the scope of the statistical analysis.

o Expected cell counts :-

⇒ We now calculate expected cell counts.

$$\text{expected count} = \frac{\text{row total} \times \text{column total}}{n}$$

EXAMPLE 9.9 Expected Cell Counts

CASE 9.2 What is the expected count in the upper-left cell in the table for Case 9.2, bottles of French wine sold when no music is playing, under the null hypothesis that music and wine sales are independent?

The column total, the number of bottles of wine sold when no music is playing, is 84. The row total shows that 99 bottles of French wine were sold during the study. The total sales were 243. The expected cell count is therefore

$$\frac{(84)(99)}{243} = 34.222$$

Note that although any count of bottles sold must be a whole number, an expected count need not be. The expected count is the mean over many repetitions of the study, assuming no relationship.

Nine similar calculations produce this table of expected counts:

Expected counts for wine and music

	Music			
Wine	None	French	Italian	Total
French	34.222	30.556	34.222	99.000
Italian	10.716	9.568	10.716	31.000
Other	39.062	34.877	39.062	113.001
Total	84.000	75.001	84.000	243.001

⇒ Next is X^2 statistics & P-value.

The X^2 statistic is the sum of nine such terms:

$$\begin{aligned} X^2 &= \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}} \\ &= \frac{(30 - 34.222)^2}{34.222} + \frac{(39 - 30.556)^2}{30.556} + \frac{(30 - 34.222)^2}{34.222} \\ &\quad + \frac{(11 - 10.716)^2}{10.716} + \frac{(1 - 9.568)^2}{9.568} + \frac{(19 - 10.716)^2}{10.716} \\ &\quad + \frac{(43 - 39.062)^2}{39.062} + \frac{(35 - 34.877)^2}{34.877} + \frac{(35 - 39.062)^2}{39.062} \\ &= 0.5209 + 2.3337 + 0.5209 + 0.0075 + 7.6724 + 6.4038 \\ &\quad + 0.3971 + 0.0004 + 0.4223 \\ &= 18.28 \end{aligned}$$

Because there are $r = 3$ types of wine and $c = 3$ music conditions, the degrees of freedom for this statistic are

$$\text{df} = (r - 1)(c - 1) = (3 - 1)(3 - 1) = 4$$

Under the null hypothesis that music and wine sales are independent, the test statistic X^2 has a $\chi^2(4)$ distribution. To obtain the P-value, look at the df = 4 row in Table F.

	df = 4	
p	0.0025	0.001

χ^2	16.42	18.47

The calculated value $X^2 = 18.28$ lies between the critical points for probabilities 0.0025 and 0.001. The P-value is therefore between 0.0025 and 0.001. Because the expected cell counts are all large, the P-value from Table F will be quite accurate. There is strong evidence ($X^2 = 18.28$, df $= 4$, $P < 0.0025$) that the type of music being played has an effect on wine sales.

- Models for two-way tables :-

The first model: comparing several populations

Case 9.2 (wine sales in three environments) is an example of *separate and independent random samples* from each of c populations. The c columns of the two-way table represent the populations. There is a single categorical response variable, wine type. The r rows of the table correspond to the values of the response variable.

We know that the z test for comparing the two proportions of successes and the chi-square test for the 2×2 table are equivalent. The $r \times c$ table allows us to compare more than two populations or more than two categories of response, or both. In this setting, the null hypothesis "no relationship between column variable and row variable" becomes

H_0: The distribution of the response variable is the same in all c populations.

Because the response variable is categorical, its distribution just consists of the probabilities of its r values. The null hypothesis says that these probabilities (or population proportions) are the same in all c populations.

EXAMPLE 9.10 Music and Wine Sales

In the market research study of Case 9.2, we compare three populations:

Population 1: bottles of wine sold when no music is playing
Population 2: bottles of wine sold when French music is playing
Population 3: bottles of wine sold when Italian music is playing

We have three samples, of sizes 84, 75, and 84, a separate sample from each population. The null hypothesis for the chi-square test is

H_0: The proportions of each wine type sold are the same in all three populations.

The parameters of the model are the proportions of the three types of wine that would be sold in each of the three environments. There are three proportions (for French wine, Italian wine, and other wine) for each environment.

Model for Comparing Several Populations Using Two-Way Tables

Select independent SRSs from each of c populations, of sizes $n_1, n_2, \ldots, n_c$. Classify each individual in a sample according to a categorical response variable with r possible values. There are c different probability distributions, one for each population.

The null hypothesis is that the distributions of the response variable are the same in all c populations. The alternative hypothesis says that these c distributions are not all the same.

The second model: testing independence

A second model for which our analysis of $r \times c$ tables is valid is illustrated by the exclusive-territory study of Case 9.1. There, a *single* sample from a *single* population was classified according to two categorical variables.

EXAMPLE 9.11 Exclusive Territories and Franchise Success

The single population studied is

Population: recently established franchise firms

The researchers had a sample of 170 such firms. They measured two categorical variables for each firm:

Column variable: Exclusive territory? (Yes/No)
Row variable: Firm successful? (Yes/No)

The null hypothesis for the chi-square test is

H_0: The row variable and the column variable are independent.

The parameters of the model are the probabilities for each of the four possible combinations of values of the row and column variables. If the null hypothesis is true, the multiplication rule for independent events (page 287) says that these can be found as the products of outcome probabilities for each variable alone.

Model for Examining Independence in Two-Way Tables

Select an SRS of size n from a population. Measure two categorical variables for each individual.

The null hypothesis is that the row and column variables are independent. The alternative hypothesis is that the row and column variables are dependent.